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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular Examination October 2025
Third Semester-Discipline Specific Core Course (DSC)
MG3DSCZGY202-APPLIED ZOOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A.Multiple Choice Questions.
Answer all questions.Each question caries 1 marks

1. Weed fishes are **[K]/[CO3]**
a). Herbivorous fishes d). Small sized uneconomic fishes **(both are correct)**
2. Foundation sheet in hive is **[K]/[CO2]**
a) Wax sheet for comb building
3. Silkworms that produce one generation in a year are called **[K]/[CO2]**
a). Univoltine
4 In fish processing, blanching helps in **[U]/[CO3)**
c) Removing slime and reducing bacterial load
5. Which among the following is a protozoan disease affecting poultry? **[U]/[CO1)**
c) Coccidiosis
6. Diseases such as Cholera and Plague can be prevented by **[U]/[CO2]**
d) Vaccination and sanitation
(If attempted, give marks, out of syllabus)
7. Dropsy disease in fish is caused by **[K)/(CO3)**
a)Pseudomonas punctata
8. Mussels feed primarily on **[U]/[CO3]**
a). Phytoplankton
9.The recommended male to female ratio for breeding quails is **[U]/[CO1]**
b) 1:3-5
10. Due to low energy requirements, fishes can synthesize more protein, resulting in
b) Increased growth rate **[U]/[CO3]**
11. Which type of poultry feed is given to pre-layer hens between 9-18 weeks of age?
c) Developer feed **[K)/(CO1)**
12. Identify the fish used for mosquito control **[K)/(CO2)**
b). Guppy **[1x12-12]**

Part B. Short Answer Questions.
Answer any 6 questions. Each question carries 3 marks

13. Differentiate between free range and cage methods in chicken farming **[U]/[CO1)**
Free range system is an old and conventional method. The birds are allowed to roam in the open area without fencing. The feeding and breeding of the birds are not taken care of. In this system, investment is nil and large number of birds are lost by predation. This method is often used for indigenous breeds.
Cage System or Battery System is a modern method in which individual birds are kept in cages which are arranged in rows and tires one above the other. Feeders and waterers must be provided outside the cage. The initial investment and the cost of replacement of the cages are comparatively high. This system is commonly used for layers to facilitate egg collection and maintenance of hygiene.The advantages of this system are low space requirement, prevent cannibalism, assessment of individual performance, easy identification of infectious birds and culling **(Any 3 differences- 3 marks)**

14. Comment on the causative organism, symptoms and control measures of Foot and Mouth Disease [U]/[CO1]

Causative organism- Foot and Mouth Disease virus
Symptoms- high fever, followed by the appearance of vesicles (blisters) in the mouth, on the tongue, gums, lips, teats and between the hooves. These blisters rupture, leading to excessive salivation, lameness, difficulty in eating and walking,weight loss and a significant drop in milk yield and productivity.
Control of FMD mainly on regular vaccination, strict quarantine measures and movement control of animals during outbreaks. Sanitation, disinfection of premises and isolation of infected animals. **(3 Points- 3 marks)**

15. Comment on algal blooms and the methods of their control [U]/[CO3]

Algal blooms are sudden outbursts of algal forms in association with heavy doses of inorganic fertilizers like urea and super phosphate. They are the result of multiplication of unicellular and filamentous algae. Algal blooms are temporary or permanent. Permanent blooms may cause a surface scum, which in turn may lead to fish mortality. **(3 Points- 1.5 marks)**

Control of Algal blooms:

- Copper sulphate 1 ppm/ha packed in cloth bags and immersed in ponds is known to eradicate algal blooms.
- Introducing the duckweed, Lemna which forms a shade over the algae and prevents their growth.
- Viruses which prevent photo assimilation of carbon dioxide can also be used.
- Introduction of algal feeding fishes like Silver carp.**(3 Points- 1.5 marks)**

16. What are the key nutritional benefits of consuming Quail meet? [U]/[CO1]

- Quail meat is very tasty, tender and good for health and recommended for children, old persons pregnant women and those attempting dieting.
- Quail meat is low in fat and cholesterol.
- This meat helps to promote body and brain development in children.
- Quail meat with low calorific value and high protein content **(3 Points- 1.5 marks)**

17. Scientific name of any two exotic egg layers [K]/[CO1]

Carassius auratus- Goldfish
Betta splendens- Siamese Fighting Fish
Pterophyllum scalare- Angel fish
Colisa lalia- Dwarf gourami
Astronotus ocellatus- Oscar **(Any 2- 1.5 marks each- liberal valuation) (If chicken, quail and duck exotic breeds were written, give marks)**

18. Write a note on the advantages of consuming milk. [U]/[CO1]

- Milk is a highly nutritious, natural liquid food and is considered as a complete,balanced diet due to its rich content of essential nutrients.
- The advantages of milk include improved bone health, enhanced muscle growth, boosted immunity, better hydration and reduced risk of osteoporosis and hypertension.
- Regular milk consumption has been associated with lower risk of certain chronic conditions, such as type 2 diabetes and cardiovascular disease.**(3 Points- 3marks)**

19. Comment on Rhode Island Red breed of chicken. [U]/[CO1]

- Originated from Rhode Island in new England, they are popular in England, Canada and United States.
- Rhode Island Red has a rich reddish brown plumage which is well glossed.
- The fowls have a long rectangular and broad body with a deep breast. Legs and feet are well developed with yellow and brown colour.
- It is a dual purpose breed, famous for its high egg production (about 200–250 brown eggs per year) and excellent meat quality.
- These birds are robust, adaptable, and disease resistant and easy to manage.
- Rhode Island Reds are often used in crossbreeding programs to improve productivity in indigenous and hybrid flocks. **(Any 3 Points- 3 marks)**

20. What are the characteristic symptoms of rice bug infestation? [U]/[CO2]

The nymph and adult animal feed on the sap of the rice plant, settles on the paddy plant just at the time when the flowering begins. The nymphs and young bugs mainly feed on the milky grains. The empty grains turn whitish and show a puncture mark. The characteristic damage is called **chaffy grains.** Yellow spots appear on the leaves due to excessive sucking. Both the plant body tissue and seeds are destroyed in large scale by this pest.

[3x6-18]

Part C. Short Essay Questions.
Answer any 4 questions.Each question 5 marks

21. What is artificial insemination? Comment on their advantages [U]/[CO1]

Artificial insemination is a reproductive technology used in animal breeding in which semen from a superior quality bull is collected, processed, stored in liquid nitrogen and artificially introduced into the reproductive tract of a female cow during estrus to achieve fertilization.

Advantages

• Artificial insemination allows the use of semen from genetically superior bulls resulting in better quality offspring and improvement in the genetic features of the cattle herd.

• This method also helps to overcome challenges posed by geographical distances or lack of suitable breeding males in remote rural areas. Semen of exotic breeds can be collected and transported to inseminate indigenous breeds promoting cross breeding.

• The semen obtained from a single ejaculate can be used to fertilize many cows and the semen of any breed can be introduced into the cow for conception

• Artificial insemination ensures better planning and synchronization of breeding activities.

• Even old and injured sires can be used for collecting semen. Storage of sperms is possible for many years and the sperms of desirable traits are available for many years even after the bull has died.

• Farmers can breed their cows with the best available bulls without maintaining a male on the farm, saving the cost and botherations involved in maintaining bulls.

• This method is followed under strict hygiene and good sanitary conditions, reduces the risks associated with natural breeding, including the spread of sexually transmitted diseases and injuries to animals**.**
(Definition-1 mark; Any 4 advantages-4 marks)

22. Discuss the characteristics and compatibility of Khaki Campbell and Indian Runner ducks as layer breeds [U]/[CO1]

Khaki Campbell is the most popular exotic breed in India and globally, reared for egg production. Originating in England, it is known for its remarkable prolific egg-laying ability, producing around 300–320 eggs per year. The name 'Khaki' refers to the fawn or light brown color of its plumage, which resembles military khaki uniforms. Khaki Campbell ducks are small to medium-sized, a straight neck and an upright stance. It weighs around 1.6 to 2.2 kg and is known for its early maturity. The eggs are white to pale green in color and are rich in nutrients. These ducks are hardy and adapt well to various climates, including the tropical conditions of India. They are non-broody, meaning they rarely sit on their eggs, which allows for continuous egg production. They are also good foragers and thrive in semi-intensive and extensive systems, feeding on insects, snails and weeds. Due to their calm nature and high egg yield, they are ideal for both small-scale rural farmers and commercial duck farms.

Indian Runner is a unique and highly productive indigenous duck breed of India, well-known for its egg-laying capacity. Indian Runners have distinctive upright posture stand tall like a penguin and run rather than waddle, so they got their name. This breed originated in the Indian subcontinent Indian Runner ducks are lightweight, slender, and have long necks and legs. They come in various colors such as white, fawn and black. These ducks are excellent layers, producing around 250 to 300 eggs annually and are considered one of the best indigenous layer duck breeds.They are particularly suited for low-input, free-range and wetland farming systems due to their excellent foraging abilities. Indian Runners feed on insects, snails and leftover grains, making them cost-effective to raise. Due to their adaptability, disease resistance and high egg output, Indian Runners are an ideal choice for integrated farming systems and rural backyard poultry, especially in rice-growing regions where they help control pests naturally. **(2.5 marks each)**

23. Outline the preventive measures for dysentery [U]/[C02]

Preventive measures include maintaining good personal hygiene, using clean drinking water and ensuring proper sanitation. These are several ways to prevent these illnesses

• Wash hands before and after preparing food or eating.

• High standards of personal hygiene as well as local hygiene including sanitation must be practiced in order to prevent the disease from spreading.

• Flush or discard any stool in the toilet and clean surrounding area using hot water and detergent.

• Wash hands thoroughly after using the toilet, changing a diaper or cleaning up a child who has used the toilet, before and after attending to someone who is ill with diarrhoea.

• Wash hands after handling animals, contact with animal quarters, cleaning up animal faeces or gardening.

• Drink properly treated water. Do not swallow water while swimming in swimming pools, hot tubs or any other water bodies. **(Any 5 Points - 5 marks)**

24. Discuss the steps involved and the significance of induced breeding in aquaculture [A]/[CO3]

Hypophysation is the technique of induced breeding in fishes by using pituitary extract.

- **Selection of Breeders**
- **Collection of Pituitary Gland**
- **Preservation and Storage of Glands**
- **Preparation of Pituitary Extract**
- **Injection of Pituitary extracts to Breeders**
- **Spawning in Breeding Hapa (Brief explanation-3 marks)**

Significance of induced breeding

• It assures timely supply of seeds for fish culture and can meet the demand for any quantity of pure spawns of a particular species.

• Many fishes attain maturity in confined waters, but do not breed. Induced breeding can cut short the cost of holding potential spawners over long periods.

• The technique is very simple and can be learnt without much training.

• The expenditure of induced breeding is very low compared collection from natural resources. (**Significance-2 marks) (If induced breeding in prawn is written, give- marks)**

25. What are the nutritional and economic importance of poultry eggs? [K]/[CO1]

Nutritional Importance:

• Chicken eggs are often called a 'complete food' because they contain almost all essential nutrients required by the body.

• Chicken egg provides around 6–7 grams of protein, containing all nine essential amino acids.

• It is rich in vitamins such as A, D, E, B2 (riboflavin), B6, B12, and minerals like iron, phosphorus,calcium, selenium and zinc.

• The yolk also contains healthy fats, including omega-3 fatty acids and choline, which is important for brain development and liver function.

• They are affordable and easily digestible, making them ideal for all age groups including children, pregnant women and the elderly.

• Regular egg consumption helps to reduce malnutrition and protein deficiency in developing countries like India. **(Any 5 Points - 0.5 marks each)**

Economic Importance:

• Egg production provides a steady source of income for farmers, especially small house holds and women in rural areas.

• Inedible eggs and shells are used in the preparation of animal foods as mineral mixtures and fertilizers

• Fertile eggs are used for hatching and for the preparation of vaccines.

• Egg white is used in the manufacture of pharmaceuticals, paints, varnishes, adhesives and printing ink. It is also used in book binding,clarification of wine, tanning leather, textile dyeing and in photography.

• Egg yolk is used in the manufacture of soaps, paints and shampoos.

• Surplus egg production contributes to national food security and export potential. **(Any 5 Points - 0.5 marks each)**

26. What is composite fish culture? Name three species commonly reared in composite fish culture and mention how their distinct scological niches reduce competition [U]/[CO3]

Fish culture in which compatible and non-competing fishes are cultured simultaneously by utilizing different feeding zones of a water body so as to increase the total production from unit area is known as composite fish culture. It is also known as polyculture or mixed farming.

The best combination in India in a polyculture system is Rohu, Catla, Mrigal. Catla is a surface feeder and feeds on only Zooplankton.Rohu is a column feeder and utilizes the plankton of that area only. Mrigal is bottom feeder and feed on the plankton which is available at the bottom, mostly benthos.

OR Common carp, Silver carp and Grass carp. Common carp is bottom feeder, eats the detritus only. Silver carp is a surface feeder, but feeds only on phytoplankton. Grass carp feed only on aquatic vegetation.)

Fishes with complementary feeding habits helps to utilize efficiently the natural food present at different ecological niches in the pond for maximum fish production. The fishes never face any competition for space and food and grow well.

(Definition-1.5 marks; name of 3 species-1.5 marks; explanation -2marks)

[5x4=20]